Seeed Studio XIAO ESP32C6 - Full Pinout & Bench Wiring Card

Ultra-Compact Search & Rescue LoRa Node • SX1262 + NEO-6M + MPU6050 + SOS Button + Alert Beacon

Physical Header Layout (Top View, USB-C Port at Top)

LEFT HEADER (Pins 1 - 7)

D0	GPIO 0	Buzzer (+) / Status LED	ALERT
D1	GPIO 1	SX1262 BUSY Pin	STATUS
D2	GPIO 2	SX1262 DIO1 (Interrupt)	IRQ
D3	GPIO 21	SX1262 NSS / CS	SPI CS
D4	GPIO 22	MPU6050 SDA (Data)	I2C SDA
D5	GPIO 23	MPU6050 SCL (Clock)	I2C SCL
3V3	POWER	+3.3V Power Out / Rail	3.3V VCC

RIGHT HEADER (Pins 14 - 8)

5V	VBUS	+5V from USB-C Port	5V OUT
GND	GROUND	System Common Ground	GND
D10	GPIO 18	SX1262 MOSI	SPI MOSI
D9	GPIO 20	SX1262 MISO	SPI MISO
D8	GPIO 19	SX1262 SCK (Clock)	SPI SCK
D7	GPIO 17	NEO-6M GPS TX → MCU RX	UART RX
D6	GPIO 16	Tactile Waterproof Button	SOS BTN

Comprehensive Workbench Hookup Matrix

XIAO Pin	ESP32-C6 GPIO	Target Component	Module Terminal	Bus Type	Electrical Level	Bench Soldering / Wiring Notes
D0	GPIO 0	3V SMD Buzzer / LED	Positive (+) / Anode	Digital Out	3.3V Active HIGH	Wire Buzzer (+) and LED Anode to D0. Cathodes connect to GND rail.
D1	GPIO 1	IndieSemiC SX1262	BUSY	Digital In	3.3V Logic	Transceiver state monitoring line. High during RF state transitions.
D2	GPIO 2	IndieSemiC SX1262	DIO1	Interrupt In	3.3V Logic	Packet received (RxDone) / packet transmitted (TxDone) hardware IRQ.
D3	GPIO 21	IndieSemiC SX1262	NSS / CS	SPI CS (Out)	3.3V Active LOW	Hardware SPI Chip Select line for LoRa radio.
D4	GPIO 22	MPU6050 Accelerometer	SDA	I2C Data	3.3V Bidirectional	I2C Data bus. Internal pull-up enabled by firmware.
D5	GPIO 23	MPU6050 Accelerometer	SCL	I2C Clock	3.3V Clock Out	I2C Clock bus (standard 400kHz Fast Mode).
D6	GPIO 16	Tactile Push Button	Terminal 1	Digital In	Active LOW	Terminal 1 to D6; Terminal 2 to GND. Internal pull-up holds line HIGH until pressed.
D7	GPIO 17	u-blox NEO-6M GPS	TX	UART RX (In)	3.3V Serial (9600)	Connect to GPS module **TX**. (Leave NEO-6M RX floating/disconnected).
D8	GPIO 19	IndieSemiC SX1262	SCK	SPI Clock	3.3V Clock Out	Master SPI Clock line.
D9	GPIO 20	IndieSemiC SX1262	MISO	SPI Data In	3.3V Serial Data	Master In Slave Out (LoRa transceiver → ESP32-C6).
D10	GPIO 18	IndieSemiC SX1262	MOSI	SPI Data Out	3.3V Serial Data	Master Out Slave In (ESP32-C6 → LoRa transceiver).
RST Pad	EN / RST	IndieSemiC SX1262	NRESET	Reset Line	Active LOW	Connect SX1262 NRESET to the underside RST solder pad. Guarantees sync reset.

XIAO Pin	ESP32-C6 GPIO	Target Component	Module Terminal	Bus Type	Electrical Level	Bench Soldering / Wiring Notes
3V3	Power Rail	SX1262, GPS, MPU	VCC	Power	+3.3V DC	Supplied from AP2112K LDO (max 600mA). Feeds all 3 peripheral modules.
GND	Ground Rail	All Modules & Return	GND	Power	0V Common	Common system ground connecting battery, TP4056, MCU, sensors, and button.

Bench Assembly Quick Tips:
• **Antenna Grounding:** Always attach the 868MHz stub antenna to the SX1262 IPEX port *before* powering up the board to prevent RF power amplifier burnout.
• **GPS Fix:** The NEO-6M requires a direct view of the sky during initial boot (cold start can take 60–90 seconds). The patch antenna ceramic side must face upward.
• **Auxiliary User Button:** The onboard physical button on the XIAO PCB is routed to `GPIO 9` and operates simultaneously with the external button on D6.